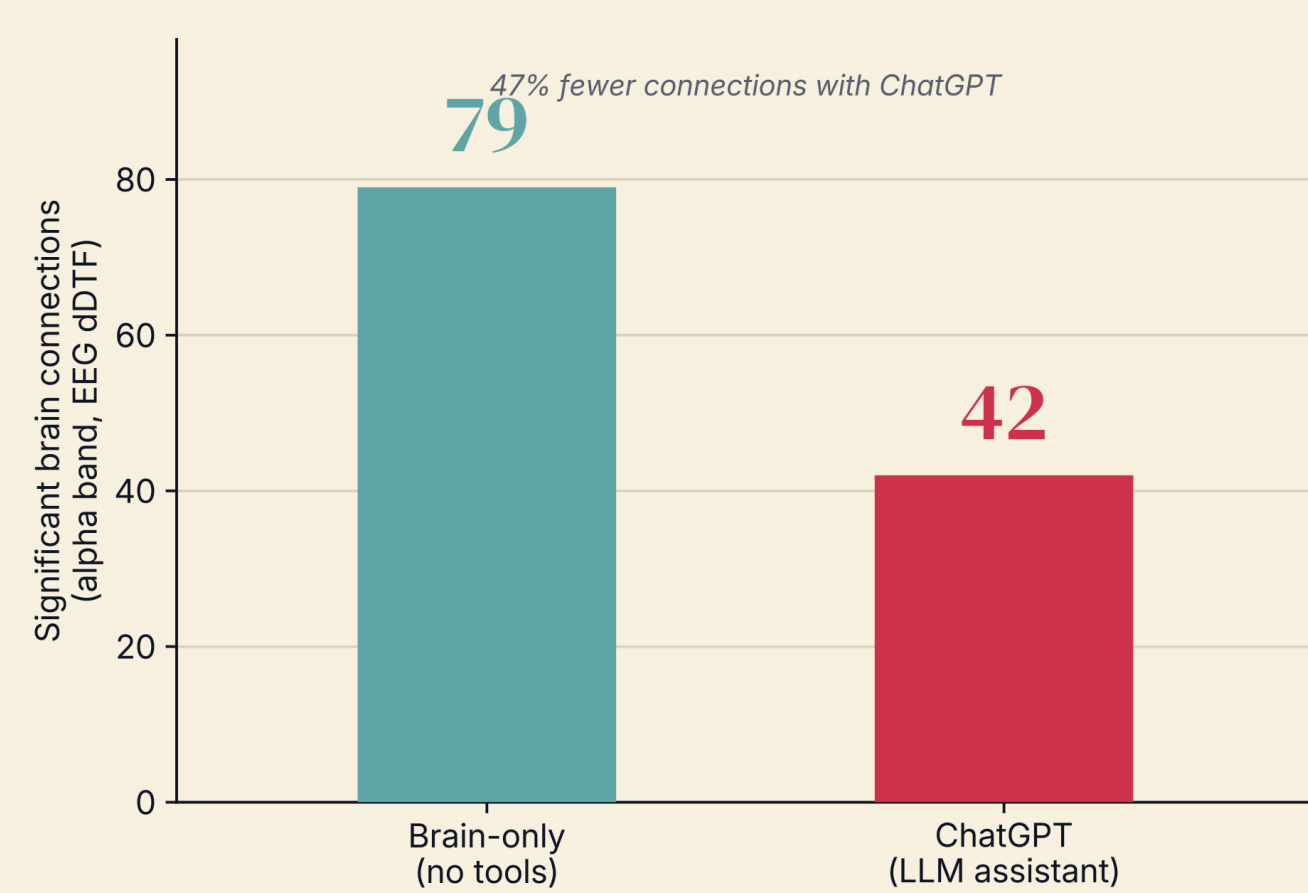


When AI thinks, the child's mind does less of it

I was on a tram when a ten-year-old told his father that ChatGPT had done his homework, and the father just chuckled, but something didn't sit right with me. Turns out this is one of the most detrimental issues in the modern age.

Cognitive Debt

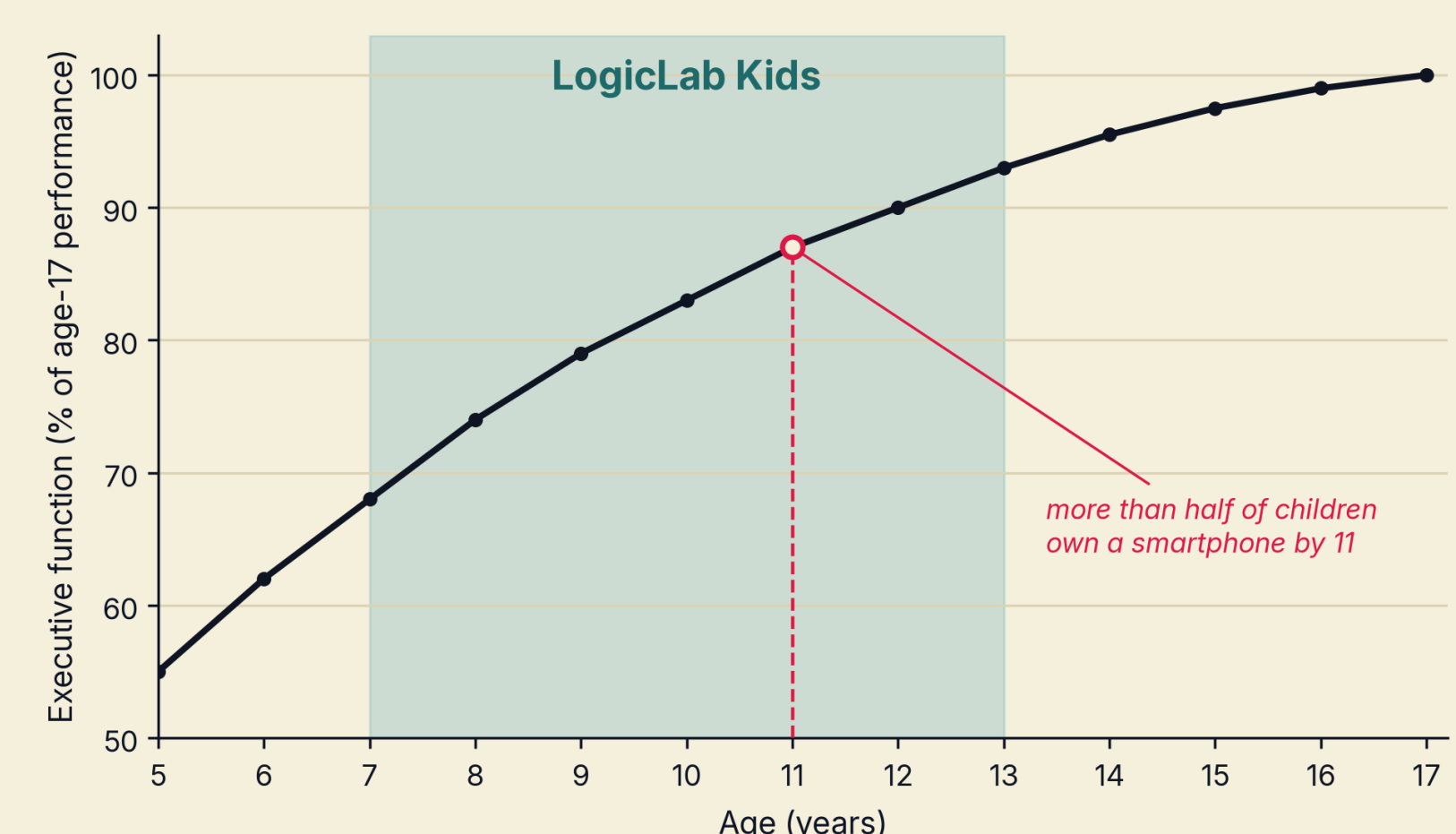
MIT did a study where they tested AI's effect on our brains. They found that people using chat gpt to write essays showcased up to 47% lower brain activity.



Kosmyna et al. (2025), MIT Media Lab, arXiv:2506.08872. Adult sample. Alpha-band dDTF, p < .05, Sessions 1-3, N = 54. Figure: own work.

The Critical Window

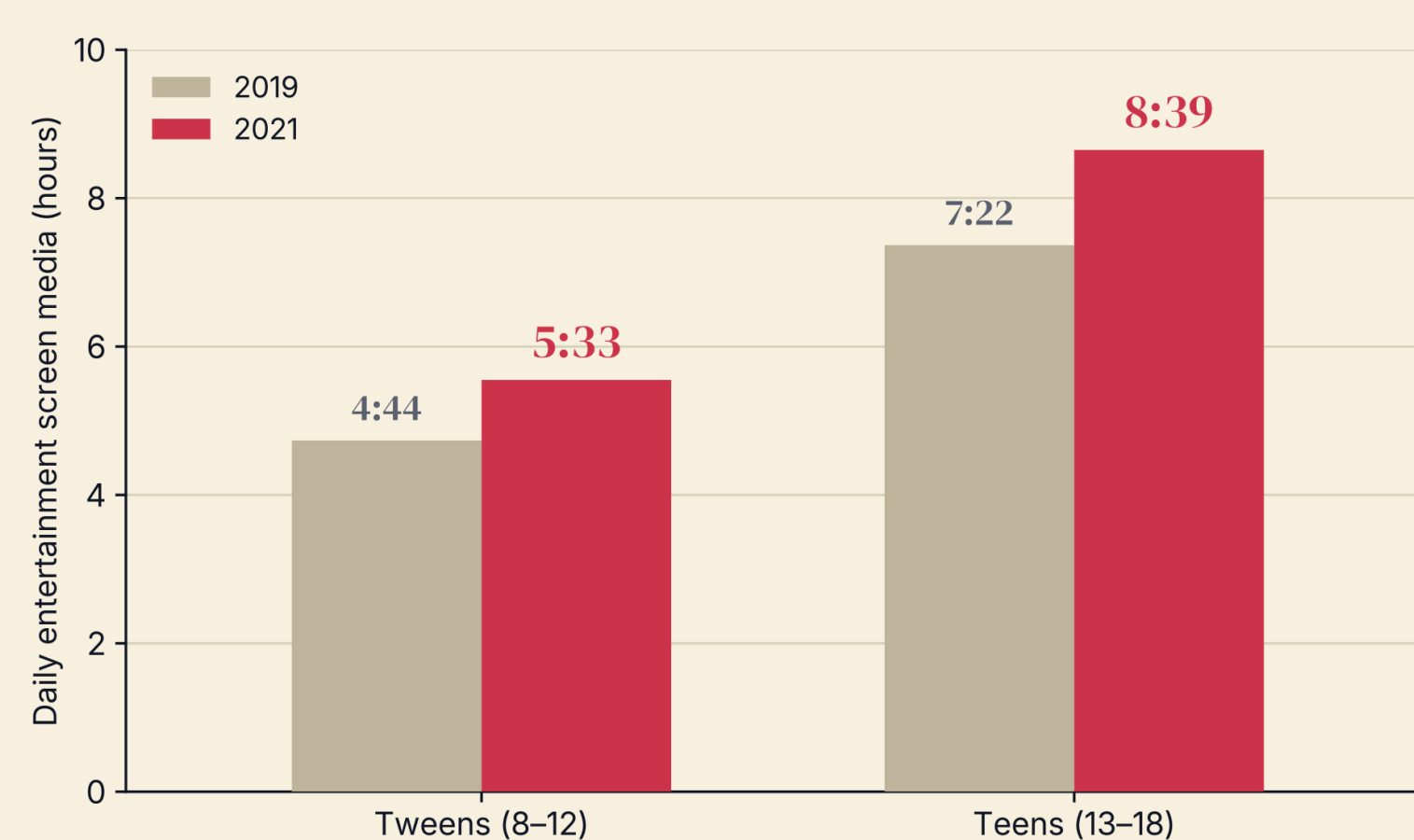
Executive function develops fastest between **ages 7 and 13**; like a muscle, if it isn't worked then, it's significantly harder to build later. And in the US, **half of children own a smartphone by 11**.



Brydges et al. (2014), Diamond (2013), Annual Review of Psychology, CAS complex-EF, N = 2,036, ages 5-17. Figure: own work.

The Collision

Children aged 8–12 spend **5 h 33 min a day** on entertainment screens, teens 8 h 39 min, up 17% in two years. Getting the childs attention is harder then ever.



Common Sense Media (2019, 2021), The Common Sense Census. +17% across both groups in two years. Figure: own work.

What I set out to build and test

Pedagogical. Make critical thinking and reasoning learnable in a fun way, which the child would actually want to do. The child builds an argument, finds the broken step, weighs evidence.

Technical. A native app where for each completed lesson the child gets screentime which the parent controls.

Research. Test against a control whether reasoning measurably improves, and whether the reward erodes the motivation it is meant to build; publish whatever the result.

The hypothesis: make thinking the price of screen time

Premack (1959): a higher-probability behaviour can reinforce a lower-probability one. Screen entertainment is a child's highest-probability behaviour and reasoning practice among the lowest, so finishing a lesson earns minutes that unlock the apps the parent chose, and **iOS, not the app, holds the lock**. So the child can't hack it.

REFERENCES

[1] Kosmyna et al. (2025). Your Brain on ChatGPT: Accumulation of Cognitive Debt. *MIT Media Lab*, arXiv:2506.08872.

[3] Common Sense Media (2019; 2021). The Common Sense Census: Media Use by Tweens and Teens.

[2] Diamond (2013). Executive functions. *Annu. Rev. Psychol.*; Brydges et al. (2014), EF trajectory ages 5–17.

[4] Premack (1959). Toward empirical behavior laws. *Psychol. Rev.*; Deci, Koestner & Ryan (1999), overjustification meta-analysis.